

Week 12: Compositions of function

Defn: Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be fxn.

We define the composition of g and f to be

the fxn

$g \circ f: A \rightarrow C$ defined by

$$(1_{\text{circ}}) \quad a \mapsto (g \circ f)(a) \stackrel{\text{def}}{=} g(f(a)).$$

ALI notation $gf = \underline{g \circ f}$

Any time you do a proof of compositions, use the defn

Example:

$$f: \mathbb{R} \rightarrow \mathbb{R} \\ x \mapsto x^2$$

$$g: \mathbb{R} \rightarrow \mathbb{R} \\ x \mapsto x+1$$

$$(g \circ f)(1) = g(f(1)) = g(1) = 2$$

$$(f \circ g)(1) = f(g(1)) = f(2) = 4$$

Note: $g \circ f \neq f \circ g$ i.e., composition is not commutative.

Same domain & codomain $\nmid$ B.J

$$(g \circ f)(1) \neq (f \circ g)(1) \implies$$

Sometimes there is a "middle" formula for $g \circ f$.

$$(g \circ f)(x) = g(f(x)) = g(x^2) = (x^2) + 1$$

$$(f \circ g)(x) = f(g(x)) = f(x+1) = (x+1)^2 = x^2 + 2x + 1$$

Warning: usually $f \circ g$ and $g \circ f$ don't both make sense,

$$f: A \rightarrow B \quad g: B \rightarrow C$$

$g \circ f$ is ok but

$f \circ g$ is not defined!

$$(f \circ g)(b) = f(g(b)) \quad \text{but}$$

$$b \in B \rightsquigarrow g(b) \in C$$

But domain of $f = A$

If: $C \subseteq A$ we can fix $f|_B$.

Examples:

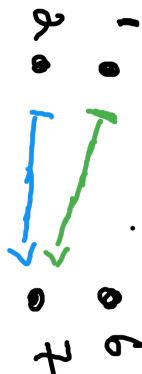
$$A \xrightarrow{f} B$$



$$B \xrightarrow{g} C$$



$$A \xrightarrow{gof} C$$



$$\begin{aligned} (gof)(1) &= g(f(1)) = g(3) = 6 \\ (gof)(2) &= g(f(2)) = g(4) = 7 \end{aligned}$$

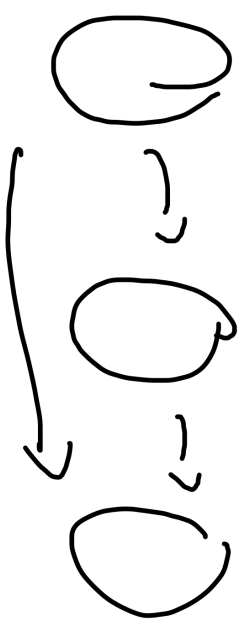
Note: f is inj + surj, but

gof is neither inj or surj

Eg $f(1) \neq f(2)$ but $f(1) = f(2)$

$$A \xrightarrow{f} B \xrightarrow{g} C$$

gof



"Simplify" (TEST Functions)

Smallest

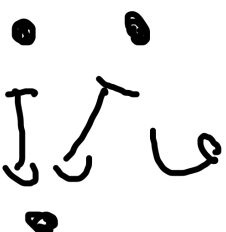
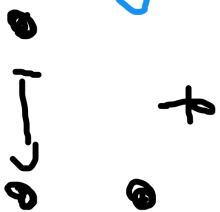
"Smallest"

inj

not

sort

fxn



sort, but not inj

fxn

got



USEFUL FOR COUNTEREXAMPLES

- (1) Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be the function $f(x) = \frac{1}{1+x^2}$ and let $g: \mathbf{R} \rightarrow \mathbf{R}$ be the function $g(x) = e^x$.
- (a) What is $(g \circ f)(0)$?
 - (b) What is $(f \circ g)(0)$?
 - (c) Give a formula for $f \circ g$ and $g \circ f$.

$$(g \circ f)(0) = g(f(0)) = g(1) = e$$

$$(f \circ g)(0) = f(g(0)) = f(1) = \frac{1}{2}$$

$$(f \circ g)(x) = f(g(x)) = f(e^x) = \frac{1}{1+(e^x)^2} = \frac{1}{1+e^{2x}}$$

$$(g \circ f)(x) = g(f(x)) = g\left(\frac{1}{1+x^2}\right) = e^{\left(\frac{1}{1+x^2}\right)}$$

(2) Let $f: \mathbf{R} \rightarrow \mathbf{Z}$ be the function $f(x) = \lfloor x \rfloor$ (i.e., round x down to the nearest integer) and let $g: \mathbf{Z} \rightarrow \mathbf{Z}$ be the function $g(n) =$ the number of distinct prime factors of n . (So $g(0) = g(1) = 0$, $g(4) = 1$, $g(6) = 2$)

(a) What is $g \circ f(\pi)$?

(b) What is $g \circ f(91.1023124)$?

(c) Is $g \circ f$ injective? Surjective?

$$f(1.1) = 1, f(1.0) = 1$$

$$f(x) = \text{the largest integer } y \text{ s.t. } y \leq x$$

$$(g \circ f)(\pi) = g(f(\pi)) = g(f(3.14159\dots)) = g(3) = 1$$

$$(g \circ f)(91.1023124) = g(f(91.1023124)) = g(91) = 2$$

$$7 \cdot 13$$

$$g \circ f \text{ not surj b/c } g \text{ is not surj, b/c } g(x) \geq 0 \quad \forall x \in \mathbf{Z}$$

$$g \circ f \text{ not inj b/c } \underline{f \text{ is not inj}}$$

$$(g \circ f)(\boxed{1}) = g(f(1)) = g(1) = 0$$

$$(g \circ f)(\boxed{1.1}) = g(f(1.1)) = g(1) = 0$$

(3) Let $f: \mathbf{Z} \rightarrow P(\mathbf{Z})$ be the function $f(n) = \{n\}$ and let $g: P(\mathbf{Z}) \rightarrow P(\mathbf{Z})$ be the function $g(S) = S \cap \{1\}$.

(a) What is $g \circ f(0)$?

(b) What is $g \circ f(1)$?

(c) Give a formula for $g \circ f$.

$$(g \circ f)(0) = g(f(0)) = g(\{0\}) = \{0\} \cap \{1\} = \emptyset$$

$$(g \circ f)(1) = g(f(1)) = g(\{1\}) = \{1\} \cap \{1\} = \{1\}$$

$$(g \circ f)(n) = g(f(n)) = g(\{n\}) = \{n\} \cap \{1\}$$

$$= \begin{cases} \{1\} & n=1 \\ \emptyset & n \neq 1 \end{cases}$$

X, Y sets

$$\text{Fun}(X, Y) \stackrel{\text{def}}{::=} \{ f: X \rightarrow Y \}$$

$$\text{Fun}(A, B) \times \text{Fun}(B, C) \xrightarrow{\quad \circ \quad} \text{Fun}(A, C)$$

$$(f, g) \longmapsto g \circ f$$

$$\text{P}(A) \times \text{P}(A) \xrightarrow{\quad \cap \quad} \text{P}(A)$$

$$(S, T) \longmapsto S \cap T$$

Lemma: Composition is associative, I.E.,

let $f: A \rightarrow B$, $g: B \rightarrow C$, $h: C \rightarrow D$ be fns. Then

Then $h \circ (g \circ f) \stackrel{(*)}{=} (h \circ g) \circ f$.

$h \circ g \circ f$

Proof: The domain + codomain agree.

(WTS: $\forall a \in A$, $(h \circ (g \circ f))(a) = ((h \circ g) \circ f)(a)$)

Let $a \in A$. Then $(h \circ (g \circ f))(a) = h((g \circ f)(a)) = h(g(f(a)))$.

Also, $((h \circ g) \circ f)(a) = (h \circ g)(f(a)) = h(g(f(a)))$.

These are equal, so $h \circ (g \circ f) = (h \circ g) \circ f$. $\square$

NOTE: to prove for 4 or more, induct, using $\rightarrow$ as base case

$$f \circ (g \circ (h \circ u)) = (f \circ g) \circ (h \circ u) \dots$$

use $\cdot, \rightarrow$ to check
 $\bullet \rightarrow \bullet \rightarrow \bullet$ truth

(4) Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be functions. Prove or disprove each of the following:

- (a) If f and g are injections, then gf is an injection.
- (b) If f and g are surjections, then gf is a surjection.
- (c) If f and g are bijections, then gf is a bijection.
- (d) If gf is an injection, then f and g are injections.
- (e) If gf is a surjection, then f and g are surjections.
- (f) If gf is a bijection, then f and g are bijections.
- (g) If gf is an injection, then f is an injection.
- (h) (HW) If gf is an injection, then g is an injection.
- (i) (HW) If gf is a surjection, then f is a surjection.
- (j) (HW) If gf is a surjection, then g is a surjection.
- (k) If gf is a bijection, then f is a bijection.
- (l) If gf is a bijection, then g is a bijection.
- (m) If gf is an injection and g is a bijection, then f is an injection.

$g \circ f$ means

$$g(x) = g(y) \Rightarrow x = y$$

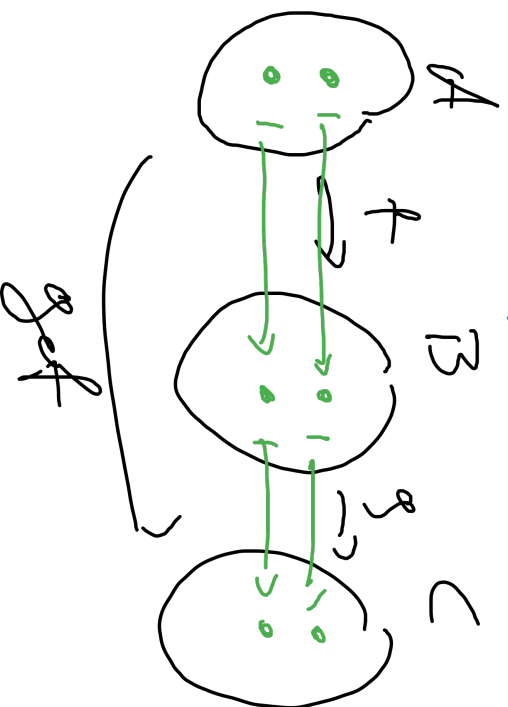
Apply w/ $x = f(a)$
 $y = f(b)$

4a, proof:

Sps f and g are inj . (wip: $g \circ f$ is inj , i.e., $\forall a, b \in A$, $(g \circ f)(a) = (g \circ f)(b) \Rightarrow a = b$)

Let $a, b \in A$. Assume $(g \circ f)(a) = (g \circ f)(b)$. Then $g(f(a)) = g(f(b))$.

Since g is inj , $f(a) = f(b)$. Since f is inj , $a = b$. $\square$



$$A \xrightarrow[\substack{f \\ g}]{\text{gof}} B \xrightarrow{g} C$$

$$(\text{gof})(A) = C$$

$$\text{If } \text{im } \text{gof} = C, \text{ If}$$

24) Sp's f and g are surj. (WTS: gof is surj. $\text{If } \forall x \in C, \exists a \in A \text{ s.t. } (\text{gof})(a) = x$)

Let $x \in C$. Since g is surj, $\exists b \in B \text{ s.t. } g(b) = x$. Since f is surj,

$$\exists a \in A \text{ s.t. } f(a) = b. \text{ Then } (\text{gof})(a) = g(f(a)) = g(b) = x. \quad \square$$

(4) Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be functions. Prove or disprove each of the following:

- ~~1~~ (a) If f and g are injections, then gf is an injection.
- ~~1~~ (b) If f and g are surjections, then gf is a surjection.
- (c) If f and g are bijections, then gf is a bijection.
- ~~X~~ (d) If gf is an injection, then f and g are injections.
- ~~X~~ (e) If gf is a surjection, then f and g are surjections.
- ~~X~~ (f) If gf is a bijection, then f and g are bijections.
- ~~1~~ (g) If gf is an injection, then f is an injection.
- (h) (HW) If gf is an injection, then g is an injection.
- (i) (HW) If gf is a surjection, then f is a surjection.
- (j) (HW) If gf is a surjection, then g is a surjection.
- (k) If gf is a bijection, then f is a bijection.
- (l) If gf is a bijection, then g is a bijection.
- (m) If gf is an injection and g is a bijection, then f is an injection.

a and $b \rightarrow c$



Counterexample to d & e,
not counter to g

Proof of g: Suppose gf is inj. Let $a, b \in A$.

Suppose $f(a) = f(b)$. Then $g(f(a)) = g(f(b))$,

and since gf is inj, $a = b$. $\square$